

brings a new constraint on the people designing their systems. They have the job of writing and building the test system. The PXI platform with LabVIEW software has been widely used as the test bench for verification and validation, and test cases are reused on the manufacturing line for product end-of-line testing," explains Mohanram.

Improvements in minimally invasive techniques as well as drug delivery technologies have resulted in advances in imaging and navigation technologies.

"This will continue to improve diagnostic accuracy and also enhance surgical capabilities. Upcoming solutions should also incorporate bioinformatics, which can result in early and faster diagnosis, better prognosis and tailored therapies. Personalisation of such devices could help a lot, where individuals' medical history and information could be available easily and at one place whenever needed," adds T. Anand, managing director, Knewron.

After a medical device is designed and developed, it has to be approved before being marketed. What's important to note is that medical device approvals are treated differently in different countries and controlled by the local authorities.

"For example, in European Union, requirements related to medical devices are illustrated in the form of directives. Medical devices, active implantable medical devices and in vitro diagnostic medical devices can be placed in European market only after they are CE marked. CE mark can be achieved by showing compliance to the applicable directives of European Union. Similarly, in USA, it is necessary to obtain the FDA clearance before placing medical devices into the market. In Canada, it is necessary to register certain medical devices with Health Canada before placing them in the market," explains Kalyan Varma, country head-products, TUV Rheinland (India).

Another aspect is the growing significance of mechatronics in the field of medicine. The application of

What test instrument customers should look for

Safety, precision and reliability of operation are the primary basis for selecting a test instrument as these are the critical part of usage. Other than that, following factors also matter.

1. Serviceability. Prompt and proper servicing is very important, as defaulting on service contracts might cause unnecessary expenditure or delays for the clients, putting them off.

2. Rate contract. A favourable rate contract is important, especially in the case of big contracts.

3. Compatibility. Compatibility with legacy instruments is a major factor for big companies as they have to ensure compatibility between the large number of equipment that they maintain. Moreover, radically new equipment might also require extensive re-training of staff handling the equipment.

4. Brand image and features. If a decision needs to be made between two products of the same price and features, consider the brand image of the company and any proprietary features that they offer.

mechatronics in medical electronics has now risen from simple positioning systems to robotic surgical devices, and even surgery simulation systems based on haptics. Performing verification and validation is one approach to reduce the errors caused by the system. In order to measure how good the device is at avoiding or recovering from errors, a term called 'mechatronics maturity' is used. Virtual test instruments are also available, which allow designers to probe complete systems for all the signals from the sensors and actuators.

Next-generation solutions

Just as electronics technology moves forward in leaps and bounds, all fields that depend on electronics also get the benefit of superior technology. Biomedical instrumentation is one area that is continuing to grow.

Anand shares a new innovation from Knewron: "We are working on design and development of an automated medicine vending machine, which perhaps is not a new concept for developed countries but has very high utility for a developing country like ours. Especially for rural areas, this could be extremely helpful. Availability of basic critical medicine and OTC drugs has been the pain area for a long time, and we believe that this solution may address the issue in a better way."

Latest advances in medical technology have also enabled patients to access personal health information in real time, helping them to maintain their

health. These devices should function perfectly at all times as the information provided by them is critical for the patient.

"A well-known Fortune 500 company that produces high-quality medical devices needed to perform comprehensive product life tests on its newest portable medical devices used for health monitoring. Specifically, they needed to measure reliability and longevity of the consumer product. The system needed to perform simultaneous tests on 30 units and generate sufficient data with the proper sample size so that design problems could be statistically differentiated from assembly problems or other anomalies"—explains a case study from National Instruments.

What's next?

Precision is of paramount importance for the highly sophisticated machines that we see today—and quality of medical care would deteriorate if test equipment fail to guarantee efficient and safe operation of electro-medical devices.

"Upcoming solutions in electro-medical domain should focus on cost innovation since this would be the driving force for proliferation of such devices and services based on these solutions. Besides cost innovation, miniaturisation, accessibility and possibility to interlink with many other devices would be the most desired features," adds Anand. ●

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